A number is only useful with its trail

Housing Evidence

Metamorphic testing across a bounded multi-agent system

This project asks whether a cooperating agent system preserves precise housing-trend relationships when its inputs change. It combines genuine local model decisions with exact Python tools, then tests the complete application through paired metamorphic executions.

Frozen feed	Evaluation design	Public agent boundary
34,560 observations 12 ZIPs / 24 months	10 relations x 30 bases 3 paired seeds per base	Orchestrator 5 internal child roles

The application

A user selects Dallas-themed ZIPs, property types, bedroom, price and size bounds, and a twelve-month window. Price and Supply inspect the same immutable selection. Region runs when public comparison is requested or a review activates a diagnostic. Reviewer can send work upstream; Composer selects grounded facts; the parent Orchestrator alone returns the verified result.

The empirical question

For each of ten declared relations, does at least one of three fixed paired seeds produce a relation or eligible output-contract violation on a distinct base? The workload contains thirty eligible bases per relation. The unit of the primary rate is a base with its entire seed panel, not an individual model call.

Scope of the evidence

The feed is synthetic, the scenarios are designed and overlapping, and the model is one local frozen profile family. The work tests relational consistency and execution reliability. It does not estimate real Dallas prices or establish an arbitrary-query failure probability. [1,2]

Define the quantity before testing it

The output contract fixes what a twelve-month trend means before any relation is evaluated. Asking-price observations, physical homes and unique sellers are different quantities; only the first is represented here.

Snapshot SHA-256: bb0e9f3ed47a65eabbaab4e0a4de192ca7918a61eee111e273a70e145f87692b

Frozen synthetic observations

The generator emits 120 monthly observation records per ZIP for twelve ZIPs across October 2024 through September 2026. Record identity is unique; listing labels deliberately repeat across months. Counts include each qualifying observation. A repeated record identity invalidates the input rather than silently deduplicating it. Independent malformed rows are quarantined with evidence.

Exact statistics

For month t, count is the size of the qualifying observation multiset. Median list price is the median of its raw USD prices, with the exact mean of the two central values for even cardinality. Median price per square foot is the median of individual price/size ratios, not a ratio of medians. Regional price-per-area medians pool individual ratios for each ZIP.

Calendar and arithmetic

The visible window contains twelve calendar months. Historical lookbacks are retained where available. Month-over-month and year-over-year change are 100 times (current/previous minus 1), evaluated from exact values. Undefined values remain null. Integers and Fraction values carry internal arithmetic; serialization uses six decimal places with half-even rounding. Display conversion uses exactly 0.09290304 square metres per square foot.

Immutable criteria

All raw filters are inclusive. Price bounds apply before display conversion. A review may change scan strategy, diagnostics and the active specialist set, but cannot broaden the user's criteria to conceal low support. Empty months remain visible and are not plotted as zero prices.

Hand-calculated example	Price (USD)	Size (sq ft)	Individual ratio
Observation A	100	100	1
Observation B	300	50	6

Median of individual ratios = (1 + 6) / 2 = 3.5. Ratio of medians = 200 / 75 = 8/3. The contract uses 3.5. This is a mathematical example, not market data or a campaign result.

One public boundary, five internal roles

The parent uses Fast Agent agents as tools. This is distinct from the framework's native planning Orchestrator decorator. The choice was checked against official documentation, installed 0.10.45 source, and an instrumented integration probe. [5]

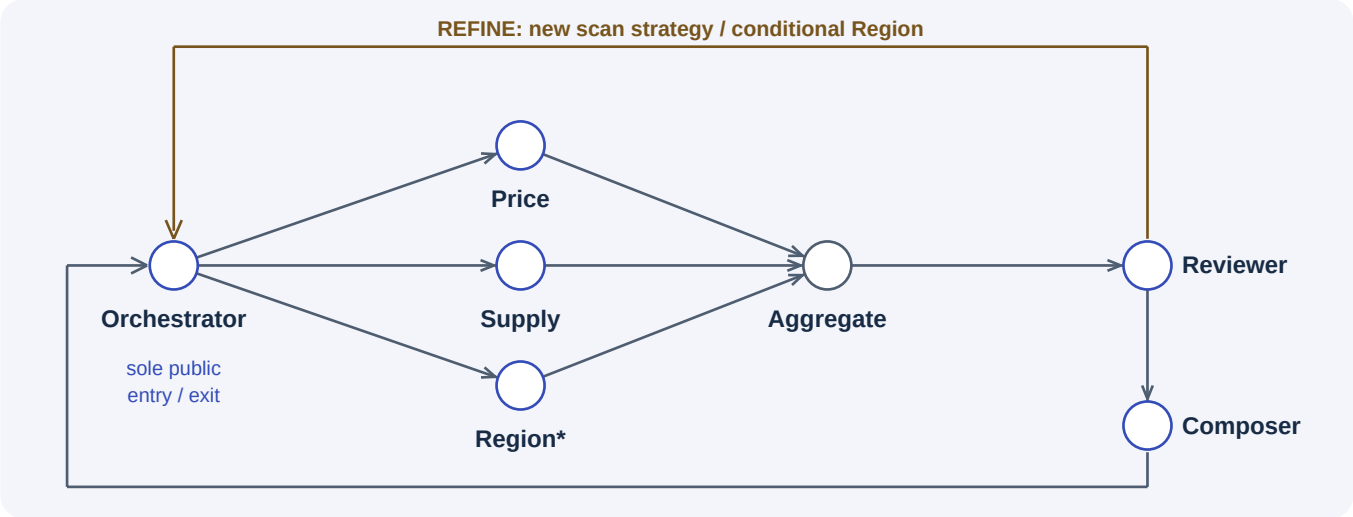

Figure 1. Arrows show evidence dependencies; every child call and return passes through the parent. Region* is conditional; Aggregate is deterministic Python. The amber refinement changes actual work.

Decisions and statistical truth

Each role performs a genuine model call and returns a role-specific strict decision. Trusted Python binds dataset, query, plan and selection handles; model text cannot supply a count, median or arbitrary tool argument. Aggregation requires validated Price and Supply receipts and the requested Region receipt. It checks their view, query, plan, month and selection identities before constructing a result.

Capability and history boundaries

Child contexts are fresh and do not inherit or merge a source conversation. The parent may delegate only the currently permitted actions. A specialist cannot call another specialist or choose arbitrary SQL or file paths. Premature review, inactive roles, foreign receipts and unapproved plan changes fail explicitly. Canonical output is published only after the real parent returns a validated final acknowledgement.

A small explicit state machine

Plan, select, scan, aggregate, review, compose and finalize are explicit states. At most two distinct semantic refinements are allowed. One schema repair is separate from one framework transport retry; underlying SDK retries are disabled. A conservative context guard rejects overflow before inference. These controls constrain decisions without replacing the model with a hidden scripted agent.

The feedback changes the work

An actual sparse source in the headline campaign triggered full rescanning and a diagnostic specialist. Counts stayed the same; the plan, receipts and acceptance decision changed.

Recorded step	Actual action / evidence
Indexed scan	Price + Supply; 35 visible / 63 with available lookbacks
Reviewer / pass 1	REFINE: SPARSE_SLICE; no index mismatch recorded
Parent replan	full_csv; Price + Supply + diagnostic Region; same criteria
Fresh scans / review	Same counts; ACCEPT_LIMITED / CONFIRMED_SPARSE
Compose / parent return	Sparse and missing-history caveats; 16 model decisions

Natural source trace: artifacts/campaigns/headline-live-v1/cases/MR01-b010-s104729/attempt-001/source/runs/20261016T164018Z-78c6924bf7/trace.jsonl. Auxiliary control: artifacts/traces/20261016T143259Z-82736e984f/trace.jsonl.

Natural headline trajectory

MR01-b010, seed 104729, source: 35 visible observations across July 2025-June 2026, with 2-5 per month. Reviewer requests EXACT_RESCAN_WITH_REGION for SPARSE_SLICE. The parent changes indexed scanning to full_csv and activates diagnostic Region. All five child roles and the parent make genuine model calls; the completed run records 16 decisions.

Why an unchanged result still matters

Fresh Price, Supply and Region receipts confirm the same 35 visible observations and 63 including available lookbacks. Reviewer changes REFINE to ACCEPT_LIMITED / CONFIRMED_SPARSE. Criteria, selection and canonical-result hashes stay constant; plan and receipt hashes change. The rescan confirms sparsity rather than repairing an index gap or relaxing the query. Composer retains sparse-support and unavailable-lookback caveats.

Evidence classes stay separate

A separate auxiliary index-omission control deliberately hides one qualifying October observation: INDEX_COVERAGE_MISMATCH causes a rescan that recovers it. That control and policy-simulated guard fixtures remain labeled auxiliary evidence. The natural trajectory above is one original campaign execution, not an extra base or a prevalence estimate.

Invariance needs a precise projection

Metamorphic testing checks a declared relationship across executions rather than demanding a known answer for each one. A relation is a hypothesis on a subdomain; an unjustified full-response equality can flag harmless metadata or miss a vacuous transformation. [2]

MR01 - Row permutation

Shuffle all raw rows with a recorded transformation seed. On a valid complete snapshot with at least two qualifying observations and changed row order, canonical month/summary/rank values must match exactly. This targets order-sensitive selection, premature stopping and unstable aggregation. Multiset statistics ignore row order; deterministic tie rules prevent accidental rank drift.

MR02 - Nominal ID relabeling

Apply bijections to record and listing identities while preserving listing equality classes and every semantic field. Canonical statistics must match exactly. This targets accidental interpretation of nominal identity or a hidden business-label deduplication rule. A bijection preserves every equality class while removing any accidental meaning of the label spelling.

MR03 - Twofold observation replication

Append a fresh-record-identity replica of every observation while retaining the same business listing labels. Every count and count delta doubles; medians, price changes and ranks stay exact. The witness proves a true doubled multiset. Unique-property metrics would require a different relation. Repeating every sorted value twice preserves its central quantile. This exposes silent deduplication or count-derived median weights.

MR04 - Display scale and area units

Switch USD to USD thousands and square feet to square metres. Canonical statistics, percentages and rankings stay exact; displayed monetary levels divide by 1000 and per-area levels also divide by 0.09290304. Display comparisons use the declared absolute 0.000001 tolerance. The source has nonzero eligible values and the display settings actually change. Pure display conversion follows filtering and aggregation. This exposes converted-unit filtering, repeated conversion and client-side statistic drift.

MR10 - Excluded-ZIP injection

Append valid fresh-identity observations in a ZIP excluded by the query, while matching its other filters. The independent witness proves that the injection could matter if ZIP filtering were absent. Canonical statistics remain exact. This targets scope leakage and incorrect predicate composition. Every injected row fails the ZIP predicate, leaving exactly the same selected multiset; the other-filter witness prevents a trivial exclusion test.

Direction and monotonicity are conditional

Changing membership does not imply a median must move monotonically. The catalog distinguishes count inclusion, scale equivariance, calendar overlap and the bounded effect of a single-month additive shift.

MR05 - Broaden the price band

Widen finite bounds from [a,b] to [max(0,a-10000000), b+30000000] cents. Independent preflight must exhibit at least one added qualifying record. Each monthly observation count is nondecreasing. Prices and medians are deliberately outside this projection because composition changes can move them either way. Identical remaining predicates make the wider selection a superset, proving count inclusion. A violation targets missing or inconsistent price predicates.

MR06 - Restrict ZIPs

Retain a proper nonempty subset with both removed and retained eligible observations. Monthly counts cannot increase. When public regional results exist, retained-ZIP medians remain exact because their own observations are unchanged. Aggregate medians and ranks are not assumed invariant. Set intersection proves count inclusion and preserves each retained ZIP multiset. This exposes sub-region leakage and inconsistent regional pooling.

MR07 - Advance the window

Advance the end month by one with complete coverage and available lookbacks. All eleven overlapping monthly rows match exactly, including count, levels and defined changes. The rolling observation total follows the exact entering-minus-leaving-month identity. Window summaries and regional rankings are not compared across changed windows. Unchanged monthly inputs and historical lookbacks determine the same monthly row. This targets off-by-one windows, dropped history and stale caches.

MR08 - Positive price scaling

Multiply every raw price, including lookbacks and any finite price bounds, by exactly 5/4. Products must be valid integer cents and membership must be preserved. Monetary levels and regional levels scale within 0.000003; percentages agree within 0.000001. Counts, count changes, ranks and null structure remain exact. Unscaled price bounds would invalidate the relation. Positive scaling preserves order; its factor cancels in relative changes. This exposes unscaled bounds, ratio mistakes and inconsistent lookbacks.

MR09 - One-month additive shift

On an unbounded price query, add exactly USD 1000 to every raw observation in one nonempty visible month. Its median price rises by USD 1000 within 0.000001. Its individual-ratio median is at least the source plus 1000/max_sqft, allowing 0.000003 for serialization. Other monthly levels/counts stay exact. Affected percentage changes and window summaries are excluded. Each individual ratio rises by at least 1000/max_sqft; median monotonicity preserves that bound. This targets incorrect month bucketing and price updates.

Count bases, preserve the paired panel

The full frozen design contains ten slice groups crossed with three end months for each relation, yielding thirty distinct semantic source queries per MR. Each is evaluated under seeds 104729, 130363 and 155921, using the same seed for source and follow-up.

Frozen preflight: 2026-10-16T14:56:07.906314+00:00. Exact cohort and witnesses: configs/headline-cohort-v1.json.

Freeze before outcomes

Raw-CSV preflight independently checks schema, coverage, transformation determinism, membership correspondence and nonvacuity. It shares no production selector, median or comparator. A preliminary sparse size band made MR05 widening vacuous; input-only checks caught that before pilot model outcomes, and a versioned corrected cohort replaced it. Both configurations remain visible.

The estimand

Let v(i,j) indicate a relation or eligible output-contract violation on base i and paired seed j. A completed base has all three outcomes; Y(i) is their maximum. The headline rate is sum(Y)/n completed eligible bases. Three seeds give three fixed evaluation conditions, not three independent base samples. Infrastructure errors keep the base incomplete; invalid preconditions remain outside the semantic denominator.

Nominal uncertainty

The two-sided Wilson interval uses z=1.959963984540054, n bases and Decimal precision 60, then stores 24 decimal places. For zero violations in thirty bases its upper bound is about 11.35%, so zero observed failures cannot establish a sub-1% risk. The designed, overlapping slices do not meet a simple random-independent sampling interpretation; the interval is explicitly nominal for this workload. [3,4]

Logical and physical work

Only successful sources enter the complete semantic input/model/seed cache; failures remain recorded and are not reused. Later relations can inherit a prior successful trajectory. Every follow-up executes afresh, with explicit source lineage. Logical pairs, physical analyses, model decisions, provider telemetry and elapsed time remain distinct. Rankings describe this declared execution policy, not isolated transformation effects.

Wilson score interval on bases	Formula
Denominator	d = 1 + z^2/n; p = x/n
Centre	c = (p + z^2/(2n)) / d
Half-width	h = z/d * sqrt(p(1-p)/n + z^2/(4n^2))
Limits	[max(0,c-h), min(1,c+h)]; n=0 gives no interval

Results from the frozen workload

The full live workload is captured. This working table reports its recorded outcomes; full offline playback and final verification are pending. The unit is a base with any violation across the three fixed paired seeds.

MR / family	n	x/n	Rate	Nominal 95% CI	Pairs*	C-faults	Inv.	Err.
MR01 / I	30	12/30	40.00%	24.59% - 57.68%	20/90	20	0	0
MR02 / I	30	12/30	40.00%	24.59% - 57.68%	22/90	22	0	0
MR03 / I	30	16/30	53.33%	36.14% - 69.77%	26/90	26	0	0
MR04 / I	30	10/30	33.33%	19.23% - 51.22%	14/90	14	0	0
MR05 / M	30	15/30	50.00%	33.15% - 66.85%	19/90	19	0	0
MR06 / M	30	12/30	40.00%	24.59% - 57.68%	23/90	23	0	0
MR07 / D	30	14/30	46.67%	30.23% - 63.86%	24/90	24	0	0
MR08 / D	30	14/30	46.67%	30.23% - 63.86%	21/90	21	0	0
MR09 / M	30	14/30	46.67%	30.23% - 63.86%	22/90	22	0	0
MR10 / I	30	12/30	40.00%	24.59% - 57.68%	19/90	19	0	0

I=invariance; M=monotonic; D=directional. Pairs*=violating/completed seed pairs; C-faults=output-contract subset. All rows: seeds 104729, 130363, 155921. Source: headline-live-v1/headline.json.

Leg-record accounting: 900 logical comparisons; 1,196 recorded analysis attempts; 10,753 model inputs; 109 refinements; 270 repair requests. Sum of recorded analysis durations: 83971.07 seconds. These totals exclude interrupted prefixes without leg records; inclusive response telemetry appears on page 10.

Base any-seed rate / nominal 95% interval

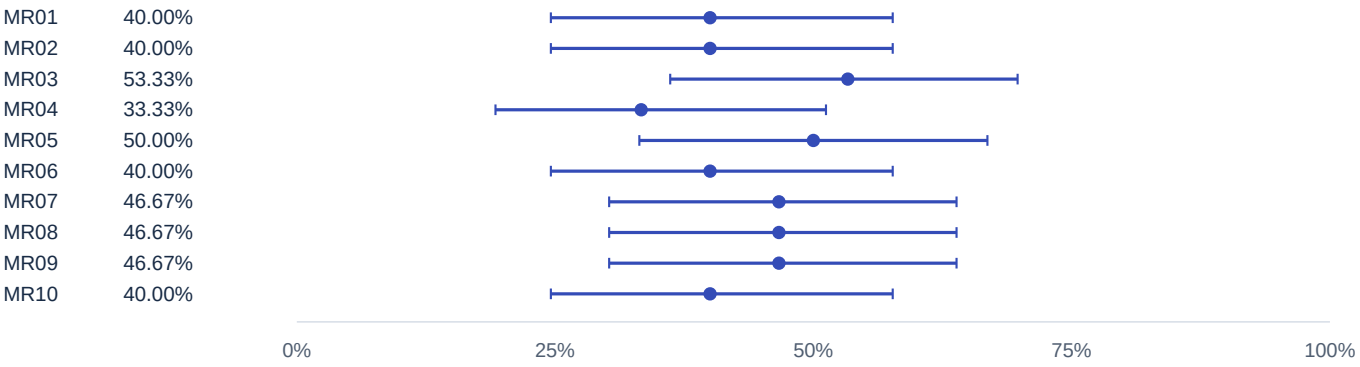

Figure 2. Nominal Wilson intervals on completed bases. Overlapping designed slices limit generalization.

Diagnose the first divergence

Natural violations are ranked by primary base-level rate, then violating-base count, then MR identifier. Diagnosis follows actual failing evidence rather than choosing an attractive story.

Observed diagnosis

All 690 numerically comparable pairs satisfy their projections; 210 pairs fail output contracts. The top three relations share one demonstrated representative mechanism: parent purpose leakage followed by a queued-role/pending-repair conflict. These are three ranked relations, not three independent bugs or evidence of transformation causality.

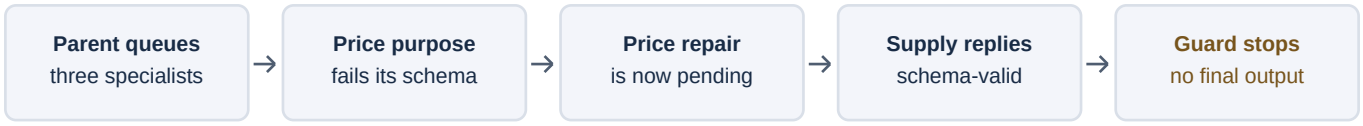

Figure 3. Observed in the three selected representatives. Supply made a real model call; the pending Price repair blocks acceptance of its valid reply.

MR03: replicated observations - 16/30 bases

b001/104729 follow-up: raw checks prove 34,560 to 69,120 unique-record rows and 2,880 to 5,760 selected observations. The parent sends public_comparison to Price, whose schema requires statistics. Price fails; Supply returns valid JSON while Price repair is pending. The controller stops before numeric tools. The source reuses MR01's successful capture.

MR05: wider price band - 15/30 bases

b001/104729 source fails through the same four-decision path. Independent inclusive predicates prove 2,149 source observations and a proper 2,880-observation superset after widening. Crucially, the failure precedes the transformation. It supports a coordination/contract diagnosis, not a counterexample to count monotonicity or a claim that widening caused failure.

MR07: shifted window - 14/30 bases

b001/130363 follow-up repeats the mechanism. All eleven overlap months have identical raw multisets; 240 records leave and 240 enter. Its source is reused from MR01. MR07 wins its tie with MR08/MR09 by identifier; overlapping intervals and shared sources prevent an isolated-sensitivity interpretation.

Reduction, alternatives and prospective correction

Each failure replays from only a manifest, terminal and four cassettes: four decisions, zero network, no saved tools or output. Dataset/semantic identities remain fixed. Valid witnesses exclude the proposed data/relation explanations for these boundaries. Other reviewed natural failures include Reviewer schema echo and Composer caveat omission. Role-specific purpose hints and queue-aware repair are proposed future fixes, not evaluated improvements.

Reproduction executes the system again

Live captures sample model behavior. Offline playback reuses those recorded decisions while rerunning the current tools, state transitions, comparisons and uncertainty calculations.

Exact semantic matching

Each turn matches prompt text, strict schema, tools, model profile, seed, iteration, dataset view and upstream receipts. Declared volatile message timestamps and framework timing/usage channels are excluded from semantic matching; raw bytes remain hashed. A miss raises REPLAY_MISS, with no approximate lookup or network fallback.

Playback is not a saved-answer shortcut

Fast Agent PlaybackLLM supplies captured decisions; the current dispatcher reruns numeric tools and receipt checks. Changed receipts invalidate the next model boundary. Integration tests replay with saved final outputs, numeric results and source traces removed. Playback blocks network connections.

Three replay mistakes caught by evidence

Sorted JSON keys reordered native tool-call identifiers; explicit ordered identifiers now preserve execution order. Repeated detached names such as reviewer[1] also replayed the first review twice; exact second-pass playback required one-decision packets. The full clean-checkout run then rejected failed-leg seed metadata: the old runner inferred it from newly written cassettes, which playback does not create. A versioned replay-only adapter now attributes the historical seed only after complete exact playback, preserving the uncorrected record. Every failed attempt remains visible.

Environment and observability

Python 3.15.0 and Fast Agent 0.10.45 are locked. Three Qwen3.5 profile defaults, digests and seeds are checked; hardware repeatability remains best effort. Offline playback needs no model server. Live inference uses a local compatible endpoint and dummy key. [6] Local Docker Langfuse 4.63.0 displays delegation, receipts, repairs and feedback through OTLP. Exact readback and browser inspection bind the dashboard to raw captures. Its versioned projection explains later-pass repairs while preserving raw tags. [7]

Local inference configuration	Frozen values
Model	Qwen3.5:9b; resolved 9.7B, Q4_K_M; Ollama 0.40.4
Sampling defaults	temperature 0.2; top_p 0.95; top_k 20; presence_penalty 1.5
Context / completion	65,536 / 1,024; reasoning_effort none; three seeded v4 profiles

Live telemetry: 10764 reported provider attempts across 10763 usage records; 28767341 prompt tokens; 720673 completion tokens. 0 responses lack attempt telemetry. Includes 10763 saved responses across 1198 physical runs, including 2 declared interrupted captures. Usage for 2 unanswered inputs is unknown. Token figures sum reported values; unrecorded retry/in-flight usage remains unknown. No monetary cost assigned. Raw audit: reports/data/headline-provider-usage.json.

What the evidence cannot establish

A precise and reproducible workload still has boundaries. These limitations determine the claims we can make, and the AI-use reflection distinguishes plausible authored material from observed execution.

Threats to validity

Construct: observation counts are not unique homes; asking-price medians are not sale prices; first-to-last change is not year-over-year change. Internal: common helpers can share faults; independent raw preflight and hand-calculated tests reduce but do not eliminate this risk. Statistical: thirty overlapping designed slices, reuse of successful sources and three fixed seed conditions are not independent representative Bernoulli trials. Intervals are nominal; rankings mix model-contract behavior, transformations and the declared reuse policy. External: a regular synthetic feed, selected ZIPs, one local model, one machine and a frozen period do not characterize live housing markets or other agents. Playback samples no fresh behavior and does not increase the sample size. Runtime comparisons are descriptive after the reboot and additional transport hop.

AI-use reflection

An LLM could draft the architecture, strict schemas, relation hypotheses, code scaffolding, diagrams, report prose and plausible tests without running this system. The initial AI-generated handoff is substantive assistance too. Those artifacts can look convincing while containing wrong API assumptions or invalid mathematics. The observed evidence is different: exact generated bytes, real parent and specialist responses, a review changing scan strategy, complete paired outcomes, recorded failures and a clean offline rerun must come from execution. An LLM can calculate a Wilson interval; it cannot honestly invent the outcomes to which that interval is applied. During implementation, an AI-chosen sparse band made price widening vacuous, which independent input counts caught. Exact replay exposed reordered tool calls and repeated clone-name response selection. A framework documentation assumption was checked against installed source rather than trusted from prose. An AI-issued recovery command also ran after a failed preflight; it was stopped and its failed attempts retained. AI_USE.md and NOTEBOOK.md record these mistakes and how they were caught. Authored test fixtures and injected omissions remain labeled controls; they do not become empirical model findings. The interface also keeps rounded display values separate from canonical statistics. Human authorship and submission remain the team's responsibility; author identities have not been fabricated.

Execution disruptions and accounting

The Mac reboot interrupted the campaign after 743 resolved comparisons. Their saved evidence was audited intact; the temporary console was lost. Recovery restored Ollama 0.40.4 and added an exact parameter-order adapter after an overly literal startup check rejected reordered, unchanged defaults. Frozen inference inputs and statistics were retained. Two startup failures and two unfinished captures remain visible. Leg-record runtime totals omit the partial prefixes; provider-usage totals include their ten saved responses. Two unanswered inputs have unknown usage. RUNTIME_RECOVERY.md and its hashed evidence document the scope.

A reviewable, rerunnable submission

The repository is the evidence package: frozen input, strict contracts, editable sources, actual captures and one-command recomputation of the reported table.

Commands

make setup installs frozen Python and frontend dependencies. make data regenerates and checks the frozen bytes. make check checks Python, schemas, generated TypeScript and frontend tests/build/lint. make dev starts the local application. make evaluate-live executes the frozen live workload. make reproduce verifies the capture and recomputes the headline campaign offline. make report builds this document from evidence. make submission-check rejects incomplete or inconsistent deliverables.

Evidence index

data/frozen and data/manifests contain the snapshot and provenance. configs/headline-cohort-v1.json contains frozen queries, seeds and witnesses. artifacts/campaigns/headline-live-v1 contains logical cases and all physical attempts. artifacts/reproductions contains current offline reruns and semantic comparisons. reports/data contains derived usage and diagnosis audits. NOTEBOOK.md records commands, AI_USE.md records assistance and mistakes, and FRAMEWORK_FINDINGS.md preserves the framework probe's narrower scope.

Coursework context

The assignment deadline is 16 October 2026 and the specified team size is two. No submission hour, earlier design-note submission, or author identity is inferred. This report records the work and evidence that actually exist. [1]

References

[1] Course-provided assignment; no author attribution inferred. TSE Assignment: Metamorphic Testing of a Multi-Agent Housing-Trends Application. Repository original, pp.1-4.

[2] Sergio Segura, Gordon Fraser, Ana B. Sanchez, Antonio Ruiz-Cortes. A Survey on Metamorphic Testing. IEEE Transactions on Software Engineering 42(9), 805-824, 2016 DOI: 10.1109/TSE.2016.2532875. Primary source.

[3] Edwin B. Wilson. Probable Inference, the Law of Succession, and Statistical Inference. Journal of the American Statistical Association 22(158), 209-212, 1927 DOI: 10.1080/01621459.1927.10502953. Primary source.

[4] NIST/SEMATECH. NIST/SEMATECH e-Handbook, 7.2.4.1 Confidence intervals. Primary source.

[5] Fast Agent project. Fast Agent workflows: Agents As Tools and Orchestrator. Primary source.

[6] Ollama project. Ollama OpenAI compatibility. Primary source.

[7] Langfuse project. OpenTelemetry for LLM Observability; v4 custom ingestion. Primary source.